A growing epithelial spheroid in an extracellular matrix: the Plexus2 implementation

`prototype/ecm` · runs in `log/okuda_ECM` · 6 August 2026

1. Sets, operators, schedule

A Plexus model is sets × fields × operators × schedule. The biological hierarchy here is four entities and one field; the basement membrane is a *specialised* ECM, not the ECM:

entity	set	representation	principal operators
epithelium	`vertex`, `cell`	3D active vertex model	`shape_energy_3d`, `morphogen_growth_3d`, `di`
junction	relation on the mesh	half-edges, keyed state	`junction_myosin`, `reconnect_t1_3d`
basement membrane	`basement_membrane_particle`	MPM + explicit bonds	`..._seed/bond/remodel/bond_break`, `integr`
interstitial ECM	`mpm_particle`	MPM fibres	`ecm_seed`, `ecm_stress`, `cell_to_ecm`
(field)	`mpm_grid`	MLS-MPM background grid	`mpm_scatter/gather/grid_update`

Every particle body scatters into and gathers from the *same* grid; that sharing *is* the mechanical coupling between them, so no pairwise contact model is written. Operators carry one of seven kinds. The four dynamics kinds return a per-set delta the engine integrates once per tick; `field`, `rewire` and `structural` mutate the grid, the relation E, or the membership $|S|$ in place and *return nothing.* That last fact drives a design decision in §5.

2. The spheroid: 3D active vertex model

Cells are faces of a closed triangulated shell; cell f's volume is the wedge from the tissue centre, $v_f = \frac{1}{3}\,\mathbf{c}_f \cdot \mathbf{N}_f$. The energy is

$$E \;=\; \sum_f \Big[K_A\,(A_f - A_f^0)^2 + K_P\,(P_f - P_f^0)^2 + \tfrac{1}{2}\Gamma P_f^2 + K_V\,(v_f - v_f^0)^2 \Big] \;+\; \Lambda \sum_e m_e\,\ell_e \;+\; K_R \sum_i \big(|\mathbf{x}_i| - R_0\big)^2, \quad (1)$$

relaxed overdamped each frame, $\mathbf{x} \leftarrow \mathbf{x} - \eta\mu\,\nabla E$ with a per-step cap. The factor m_e is new (§5); with $m_e \equiv 1$ Eq. (1) is the stock Okuda energy and every prior run is bit-identical. Growth scales each cell's targets by a cumulative s_f,

$$s_f \leftarrow s_f\big(1 + \lambda(\rho + \mathrm{Hill}(a_f))\big), \qquad A_f^0 = A_f^{0,\mathrm{init}} s_f^2, \quad P_f^0 = P_f^{0,\mathrm{init}} s_f, \quad v_f^0 = v_f^{0,\mathrm{init}} s_f^3, \quad (2)$$

and `divide_3d` splits a cell once v_f passes twice its reference. From 200 cells the tissue reaches ~ 6000 and its apical radius $4.66 \to 16.6$ (arbitrary units) in 402 frames.

3. The ECM: MLS-MPM

The matrix is a material, not a set of springs, so it is solved the way materials are: as a cloud of points that carry the stuff and a background grid that does the arithmetic. Each frame the points hand their mass and momentum to the grid nodes nearest them, the grid resolves everything arriving at each node into one velocity, and the points read that velocity back and move. Nothing is ever computed between two points directly — which is why bodies that never touch can still push on one another, and why the grid spacing sets the finest thing the matrix can feel.

We seed it as fibres rather than a uniform block, because a fibrous matrix and a homogeneous one respond differently to being pushed, and the interesting behaviour — splaying, dragging, a stress front travelling outward — lives in the fibres.

4. Spheroid ↔ ECM

The two solvers cannot share a world: the matrix grid is fixed to the unit cube, the epithelium lives in a box fifty times larger, and shrinking the tissue to fit changes which term of its energy dominates and stops it growing. So they are run as two passes joined by one geometric scale, and what crosses between them is a *recorded shape* rather than a live tissue.

That recording needs somewhere to live, and this is what the `surface` Level is. It is a set of nodes, one per membrane node, each holding a fixed direction from the tissue centre and the radius the epithelium reached in that direction at that moment. It is not a mesh and carries no mechanics; it is the epithelium's outer boundary, written down. Before it existed the same information was a lookup table, which meant nothing could act on it and nothing could be attached to it — promoting it to a Level is what lets other things bind to the tissue surface as an object rather than consult it as data.

Everything outside the epithelium then reads that one surface. Matrix fibres that end up inside it are pushed back out, first softly and then, as a backstop, by being placed outside:

$$\mathbf{a}_i \;=\; k_c\,\big(R(\hat{\mathbf{u}}_i, t) - r_i\big)_+\,\hat{\mathbf{u}}_i, \quad (3)$$

and the force the fibres push back with is collected by direction and divided by the area each direction covers, giving a pressure over the surface. A cell facing a stressed matrix then grows more slowly:

$$g \;=\; f + \frac{1-f}{1+\big(P/p_{1/2}\big)^{n}}, \qquad s_{\text{now}} \leftarrow s_{\text{prev}}\,\big(1 + g\,(f_{\text{grow}}-1)\big). \tag{4}$$

Because it is the growth *rate* that is slowed and not the size directly, a small difference compounds over hundreds of frames — which is how a modest difference in stress between one direction and another can become a difference in shape.

The coupling still runs one way. The surface is a recording, so it drives the matrix and the membrane and neither reaches back, and the second pass cannot change the tissue it is reacting to. Making it mutual means letting an operator write to `surface` instead of only reading it, which is now a change to one Level rather than a change to the architecture.

5. The two new levels

Junctions. In the vertex model every cell–cell contact pulls with the same strength. Each junction now carries its own amount of myosin, which multiplies that strength and chases a target on its own timescale:

$$\Lambda \sum_e \ell_e \;\longrightarrow\; \Lambda \sum_e m_e\,\ell_e, \qquad m_e^{ss} = a\Big(1+\beta\,\big(x_e/\langle x\rangle - 1\big)\Big), \qquad \dot m_e = \big(m_e^{ss}-m_e\big)/\tau_m. \tag{5}$$

What matters is what x_e is. If it is the junction's *length*, the loop reads *longer* $\rightarrow$ *more myosin* $\rightarrow$ *shorter*, which evens the cells out. Real myosin is recruited by *tension* and accumulates on junctions that are in the act of disappearing, which is the opposite: a junction that pulls harder shortens, which makes it pull harder still, and neighbours swap. x_e is the edge tension, read from the same energy the mechanics minimises. The two versions look alike in the spread of junction lengths and differ in how often cells exchange neighbours, so that exchange rate is what we report.

Basement membrane. A sheet of nodes joined by explicit crosslinks. The crosslinks are what make it a sheet rather than dust — material points on their own have no connections, so they cannot be crosslinked and cannot tear. Each crosslink is a spring that slowly forgets having been stretched, and snaps if stretched too far; each node is tied to a fixed *direction* on the recorded surface, which is what makes it adhesion rather than mere contact, since a node allowed to slide to the nearest point would never be stretched at all:

$$\gamma\,\dot{\mathbf{x}}_i = \sum_{j\sim i}\mathbf{f}_{ij} \;+\; \mathbf{f}_i^{\text{adh}}, \qquad \mathbf{f}_{ij} = k_b\,\big(L_{ij}-\ell_{ij}^0\big)\,\hat{\mathbf{d}}_{ij}, \qquad \dot\ell_{ij}^0 = \big(L_{ij}-\ell_{ij}^0\big)/\tau_r, \tag{6}$$

$$\mathbf{f}_i^{\text{adh}} = \kappa\big(\mathbf{a}_i-\mathbf{x}_i\big), \qquad \mathbf{a}_i = \mathbf{c} + \hat{\mathbf{u}}_i\big(R(\hat{\mathbf{u}}_i,t)+\delta\big), \qquad \hat{\mathbf{u}}_i \text{ frozen at seeding,} \tag{7}$$

with a crosslink dying once $(L-\ell^0)/\ell^0 > \varepsilon_b$. Note the left-hand side of Eq. (6): the forces set a *velocity*, not an acceleration. A stretched crosslink makes its two nodes approach at a speed proportional to the stretch, and they stop the moment it is relieved. Forgetting is not optional: a sheet that only stretches cannot enclose a surface that more than triples in size. And because area grows as the square of the radius while new material is added as a fraction of what is already there, the sheet must be *secreted* as the tissue grows or it thins.

At this scale viscous drag beats inertia by about ten orders of magnitude, so $\gamma\dot{\mathbf{x}} = \mathbf{F}$ is the equation of motion: the sheet has no mass and no momentum.

What is called adhesion here is a tether to a surface, not a receptor. The dominant anchor in a real epithelium is to laminin, which is not modelled, and the sheet has no attachment to the surrounding stroma either — so this tether is the only thing holding it.

6. The two schematics

The two levels added on top of the spheroid and the matrix. Everything else about them — the configurations, the measurements, and the external audit that found much of it wanting — has moved to `spheroid_ecm_archive.tex` while the model is being corrected.

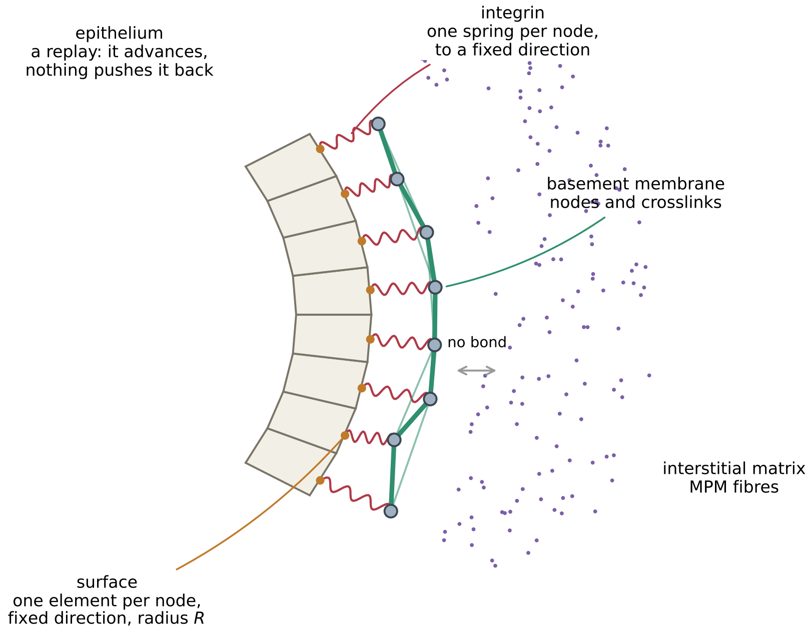

Figure 1: **What the membrane is made of.** Four objects, and the arrows between them are the whole model. The *epithelium* (cream wedges) is a recording: it advances on a prescribed schedule and nothing can push it back. Its outer boundary is the `surface` Level (orange), one element per membrane node, each holding a fixed direction $\hat{\mathbf{u}}_i$ and the radius $R(\hat{\mathbf{u}}_i, t)$ the tissue reached in that direction. The *basement membrane* is grey *nodes* joined by green *crosslinks*, sitting a distance δ outside. Each node is tied to its *own* surface element by one red *integrin* spring—one per node, to a fixed angular position, which is what makes it adhesion rather than contact. Outside is the interstitial matrix (purple), which the membrane never bonds to: they meet only through the shared grid, or in the spring-graph version, not at all.

Figure 2: **What a junction is, and the one loop that makes it interesting.** Left: two cells share an edge, and that edge *is* the junction—it belongs to neither cell, which is why its state is keyed by the unordered pair of vertices (i, j) rather than stored on a cell. Myosin m_e sits on the shared edge and sets its tension. Right: the feedback. A junction longer than average recruits more myosin (β), which pulls harder, which shortens it. That loop is the only thing here that Okuda's model cannot already do.